

Algebraic & Geometric Methods in Statistics

Outline and some illustrative examples in nonlinear statistics

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Goals

After the course, you can:

- **list topics** in algebraic statistics
- **recognize problems** in statistics that are answerable by algebraic methods
- **assess which algebraic methods** are suitable for solving a problem
- **apply basic algebraic tools** to solve a problem

Tentative course outline:

- ① **What is algebraic statistics?** An invitation / introduction / overview
- ② **Exponential families** 1.1. Statistical foundations 1.2. Underlying algebra
- ③ **Conditional independence and graphical models** 2.1. Statistical foundations 2.2. Underlying algebra
- ④ **Goodness-of-fit testing of models for discrete data** 3.1. Overview 3.2. Chromosome clusters in cancer cells 3.3. Network data 3.4. Challenges of large, sparse data sets
- ⑤ **Parameter identifiability** 4.1. Overview 4.2. Graphical models 4.3. Phylogenetics and evolutionary biology 4.4. Model selection: learning a causal graph
- ⑥ **Maximum likelihood estimation** 5.1. Introduction 5.2. Deciding existence of ML estimators 5.3. Algorithms for MLE: convex and non-convex optimization

Materials

Books and resources

Main textbook: Seth Sullivant “*Algebraic Statistics*”. It is available in the bookstore. (I will check with the library for an e-copy.)

General course syllabus is [here](#)

Homework and grade

Approximately 6-7 assignments, expect a usual weekly workload.

[Read full syllabus here.](#)

There are both “homework assignments” and “participation/worksheets” which may be completed during class sometimes. They all count toward your grade.

Project

Option A Designing a minisymposium that has a connection to the topics in this course (other related topics require instructor approval).

Option B Reading a paper, working on a small research project, or applying algebraic methods on a data set, and writing a report on it. Timeline will be determined soon; project will take place during second half of semester. Groups up to 2 students.

Communication

Feel free to discuss things on Canvas discussion boards!!

Saving this information:

I am going to link to this on Canvas!!



Student input time!



AhaSlides.com/PMQRI

Motivating example 1: Discrete Markov chain

Section 1.1. of the textbook. **Lecture on board.**

Action item: derive two polynomial equations for Markov chain model.

Let X_1, X_2, X_3 be a sequence of random variables taking values in $\Sigma = \{0, 1\}$.

- There are eight joint probabilities $p_{ijk} = P(X_1 = i, X_2 = j, X_3 = k)$ for $i, j, k \in \{0, 1\}$.
- A probability distribution associated to (X_1, X_2, X_3) corresponds to a point in $\mathbb{R}^8$.
- The sequence is a Markov chain if
$$P(X_3 = x_3 \mid X_1 = x_1, X_2 = x_2) = P(X_3 = x_3 \mid X_2 = x_2).$$

Question: When is a point in $\mathbb{R}^8$ the probability distribution associated to a Markov chain?

Conditional Probabilities in Terms of Joint Probabilities

For $i, j, k \in \{0, 1\}$, $P(X_3 = k \mid X_1 = i, X_2 = j) = \frac{p_{ijk}}{p_{ij+}}$ where

$$p_{ij+} = \sum_{k \in \{0,1\}} p_{ijk}.$$

Markov chain condition.

Using the above and comparing the conditional distributions for different values of X_1 , we have $\frac{p_{ijk}}{p_{ij+}} = \frac{p_{i'jk}}{p_{i'j+}}$ for $i \neq i'$, which implies

$$p_{ijk} p_{i'j+} = p_{i'jk} p_{ij+}.$$

Example

In the binary case ($\Sigma = \{0, 1\}$), simplifying yields the two polynomial equations: $p_{000} p_{101} - p_{001} p_{100} = 0$ and $p_{010} p_{111} - p_{011} p_{110} = 0$.

Characterization of the Model (Semialgebraic Set)

A point $p \in \mathbb{R}^8$ is the probability distribution associated to a (binary, 3-step) Markov chain **iff**:

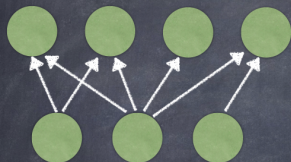
- $p_{ijk} \geq 0$ for all $i, j, k \in \{0, 1\}$,
- $\sum_{i,j,k \in \{0,1\}} p_{ijk} = 1$,
- $p_{000} p_{101} - p_{001} p_{100} = 0$,
- $p_{010} p_{111} - p_{011} p_{110} = 0$.

Therefore, the Markov chain model forms a **semialgebraic set**—the solution set of a system of polynomial equations and inequalities.

Notes and Next Steps:

- This is an example of a **conditional independence** model (future lecture)
- **Fitting the model to data:** Assume there is a true, unknown distribution p in the model generating the data. **What is p ?** (likelihood inference; future lecture)
- **Model assessment:** How well does the model fit the data? (Fisher's exact test in future lecture).

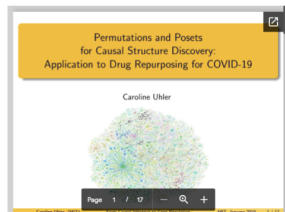
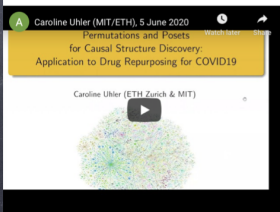
Motivating example 2: Graphical models



Algebraic Statistics
Seminar Online ->
Past Talks and
Recordings

Friday, June 5, 2020: Caroline Uhler (MIT/ETH)

- **Speaker:** [Caroline Uhler](#) (MIT/ETH)
- **Title:** Permutations and Posets for Causal Structure Discovery
- **Abstract:** Gene knockout experiments allow performing interventions in large-scale systems. This represents a unique opportunity for causal structure discovery, since it allows testing algorithms with real data and in relevant settings. We discuss the rich combinatorial, algebraic and geometric questions underlying causal structure discovery. In particular, we show that viewing causal structure discovery as an optimization problem over permutations (in the fully observed setting) or posets (in the presence of unobserved variables) can lead to algorithms with stronger consistency guarantees than previously known, which translates into better performance in terms of predicting the effect of a gene knockout experiment.

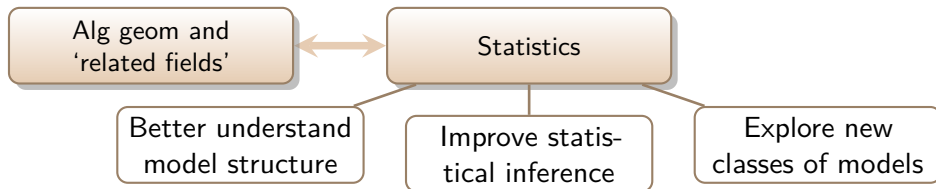


What *is* algebraic statistics?

Fact [Guiding principle]

Many statistical models are defined by (semi-)algebraic sets of parameters.

- The **geometry** of these parameter spaces determines the behavior of widely used statistical inference procedures.
- "Shape" of a statistical model: intuitive notion of fundamental importance to statistical inference.
 - E.g: is the likelihood function multimodal?
 - Does the model have singularities (is non-regular)?



Probability / statistics

- Probability distribution
- Statistical model
- (Discrete) exponential family
- Conditional inference
- Maximum likelihood estimation
- model selection
- Multivariate Gaussian model
- Phylogenetic model
- MAP estimates

Algebra/geometry

- Point
- (Semi)algebraic set
- Toric variety / ideal
- Lattice points in polytopes
- Polynomial optimization
- Geometry of singularities
- Spectrahedral geometry
- Tensor networks
- Tropical geometry

Lecture plan

We will continue now with the following topics:

- Probability Primer (Chapter 2) and
- Conditional Independence (Chapter 4)

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Appendix

Following is a 3-slide “intro” to algebraic geometry; these were slides by S. Sullivant given at a colloquium a long long time ago. They are meant to just give you a glimpse into the vocabulary. . . not to digest this immediately.

Introduction to algebraic geometry

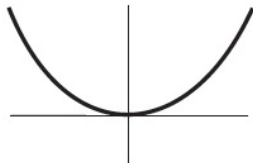
Let $\mathbb{R}[\mathbf{p}] = \mathbb{R}[p_1, \dots, p_m]$ be the set of all polynomials in indeterminates $p_1, \dots, p_m$ with real coefficients.

Definition

Let $\mathcal{F} \subset \mathbb{R}[p_1, \dots, p_m]$. The **variety** defined by $\mathcal{F}$ is the set

$$V(\mathcal{F}) := \{\mathbf{a} \in \mathbb{R}^m \mid f(\mathbf{a}) = 0 \text{ for all } f \in \mathcal{F}\}.$$

$$V(\{p_2 - p_1^2\}) =$$



Definition

Given $S \subset \mathbb{R}^m$

$$\mathcal{I}(S) := \{f \in \mathbb{R}[\mathbf{p}] \mid f(\mathbf{a}) = 0 \text{ for all } \mathbf{a} \in S\}$$

is the **vanishing ideal** of S .

Theorem (Hilbert's Basis Theorem)

Every ideal $I \subset \mathbb{R}[\mathbf{p}]$ has a finite generating set. That is, for each ideal I , there exists a finite set $\mathcal{F} \subset I$ such that $\langle \mathcal{F} \rangle = I$, where

$$\langle \mathcal{F} \rangle := \left\{ \sum h_i f_i \mid h_i \in \mathbb{R}[\mathbf{p}], f_i \in \mathcal{F} \right\}.$$

Example: Hardy-Weinberg Equilibrium

Suppose a gene has two alleles, a and A . If allele a occurs in the population with frequency θ (and A with frequency $1 - \theta$) and these alleles are in Hardy-Weinberg equilibrium, the genotype frequencies are

$$P(X = aa) = \theta^2, P(X = aA) = 2\theta(1 - \theta), P(X = AA) = (1 - \theta)^2$$

The model of Hardy-Weinberg equilibrium is the set

$$\mathcal{M} = \left\{ \left(\theta^2, 2\theta(1 - \theta), (1 - \theta)^2 \right) \mid \theta \in [0, 1] \right\} \subset \Delta_3$$

$$\mathcal{I}(\mathcal{M}) = \langle p_{aa} + p_{aA} + p_{AA} - 1, p_{aA}^2 - 4p_{aa}p_{AA} \rangle$$

